MINOR PROJECT ASSESSMENT

Cybersecurity & Networking Fundamentals

Level: Foundation Batch **Total Marks:** 100

Topics: Cybersecurity fundamentals • Networking • Network types • Internet communication • Network devices • IP addressing • IPv4/IPv6 • IP classes • Private/Public IP • Subnet masks • CIDR • Subnet calculation

Instructions

• Answer all questions unless instructed otherwise.

• Show complete working for calculations.

• Explain your reasoning in scenario-based questions.

• Use diagrams wherever useful.

SECTION A — CYBERSECURITY FUNDAMENTALS (15 MARKS)

Q1. Fundamentals (5 Marks)

Explain cybersecurity and why it is important. Explain the CIA Triad with one practical example for each component.

Q2. Security Thinking (5 Marks)

Differentiate between vulnerability, threat, risk and attack. Give one simple example for each.

Q3. Types of Security (5 Marks)

Briefly explain any five areas of cybersecurity, such as network, application, cloud, endpoint, information or physical security.

SECTION B — NETWORKING FUNDAMENTALS (20 MARKS)

Q4. What is Networking? (5 Marks)

Define a computer network. Explain why organizations use networks and describe LAN, MAN, WAN and PAN with one example each.

Q5. Network Devices (5 Marks)

Explain the purpose of a hub, switch, router, firewall and access point. Mention one major difference between a switch and a router.

Q6. How the Internet Works (10 Marks)

Explain what happens when a user enters example.com in a browser. Include DNS, IP addressing, gateway/router, TCP/IP and the web server. Draw a simple flow diagram.

SECTION C — IP ADDRESSING (20 MARKS)

Q7. IPv4 & IPv6 (5 Marks)

Differentiate IPv4 and IPv6. Explain why IPv6 was introduced and give one example address for each.

Q8. IP Classification (5 Marks)

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For 10.10.10.5, 172.16.5.20, 192.168.1.50 and 8.8.8.8, identify the IPv4 class and state private/public where applicable.

Q9. IP Components (10 Marks)

For 192.168.20.45/24, identify network address, host portion, subnet mask, broadcast address and usable host range. Show your working.

SECTION D — SUBNETTING & CALCULATIONS (25 MARKS)

Q10. Basic Subnet Calculation (10 Marks)

For 192.168.10.0/26, calculate subnet mask, number of /26 subnets from /24, usable hosts per subnet, network address, first usable, last usable and broadcast address.

Q11. Subnetting Design (15 Marks)

An organization has 192.168.10.0/24 and requires separate networks for IT—50 hosts, HR—30, Finance—20, Management—10. Design an appropriate subnetting scheme. Give Network, CIDR, Mask, Usable Range, Broadcast and Host Count.

SECTION E — SCENARIO-BASED QUESTIONS (20 MARKS)

Q12. Scenario — Company Network (10 Marks)

A company has 45 employees, 5 servers and 10 network devices. Design a basic network. State network type, required devices, role of router/switch/firewall, and explain how an employee's request reaches a website.

Q13. Scenario — IP Troubleshooting (10 Marks)

A user has 192.168.1.25/24 and can communicate with another local computer but cannot access an external website. As a junior network engineer, list checks in order, including IP configuration, gateway, DNS, connectivity testing and router role.

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MARKING SCHEME

Section	Topic	Marks
A	Cybersecurity Fundamentals	15
B	Networking Fundamentals	20
C	IP Addressing	20
D	Subnetting & Calculations	25
E	Scenario-Based Questions	20
TOTAL		100

Expected Submission

- Complete answers
- Subnet calculations with working
- Network diagrams where required
- Scenario answers with reasoning
- Clear and technically correct terminology

Instructor Note: The assessment tests conceptual understanding, subnet calculation, networking communication, and basic troubleshooting ability.

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